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Pune**

Subject: Computer Vision Laboratory

Experiment - 12

Aim: To detect and segment moving objects in video streams.

Software Tools Required: VS Code, Python 3.12

Theory:

Detecting and segmenting moving objects in video streams is a fundamental task in computer vision, widely used in applications such as traffic monitoring, surveillance, and autonomous systems. The main goal is to identify regions in a video where motion occurs and separate these moving objects from the static background.

1. Motion Detection

Motion detection relies on analyzing the changes between consecutive frames of a video. Areas where pixel values change significantly indicate movement. This is typically done using frame differencing, which compares each frame with the previous one to highlight changes.

2. Preprocessing

Before detecting motion, the video frames are preprocessed to improve accuracy:

- **Grayscale Conversion:** Converts color images to grayscale, reducing computational complexity while preserving essential information for motion detection.
- **Noise Reduction:** Techniques such as Gaussian blur are applied to smooth out minor variations in pixel intensity, which helps prevent false motion detection caused by camera noise or small lighting changes.

3. Thresholding

After preprocessing, a thresholding operation converts the difference image into a binary image, where:



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- White regions represent motion (foreground).
- Black regions represent static areas (background).

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This simplifies the identification of moving objects.

4. Contour Detection and Segmentation

- Contours are extracted from the binary image to identify connected regions corresponding to moving objects.
- Small or irrelevant contours are filtered out to remove noise.
- Bounding boxes are drawn around significant contours, effectively segmenting the moving objects for visualization and further analysis.

5. Frame Update and Continuity

The algorithm continuously updates the frames, ensuring motion is detected across the entire video stream. This allows real-time tracking of moving objects as the video progresses.

Applications

This approach is widely used in:

- Traffic flow monitoring and vehicle detection.
- Security surveillance and intrusion detection.
- Object tracking in robotics and autonomous systems.

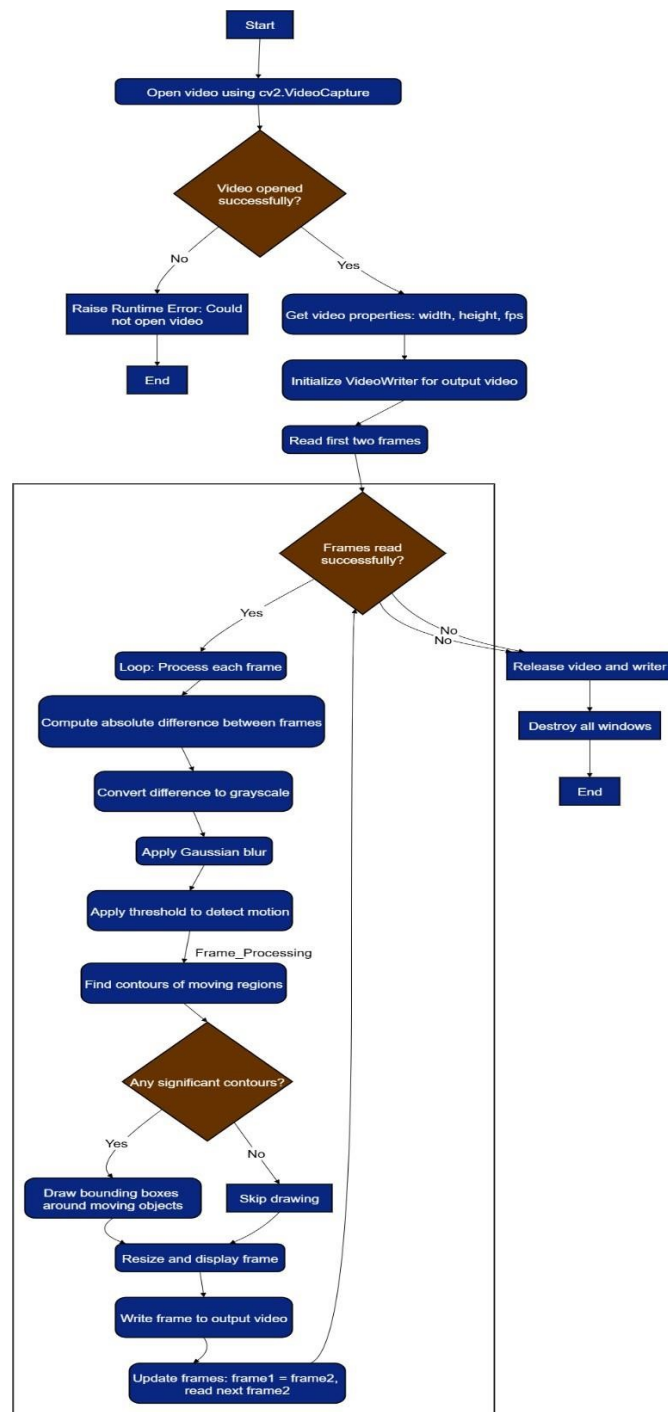
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Flowchart



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Part-1





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Program with Output:

Program:

```
import cv2

# Open the video
cap = cv2.VideoCapture('traffic.mp4')

# Get video properties
frame_width = int(cap.get(cv2.CAP_PROP_FRAME_WIDTH))
frame_height = int(cap.get(cv2.CAP_PROP_FRAME_HEIGHT))
fps = cap.get(cv2.CAP_PROP_FPS)

# Define the codec and create VideoWriter object
out = cv2.VideoWriter('output_motion.avi', cv2.VideoWriter_fourcc(*'XVID'), fps, (frame_width, frame_height))

# Read two initial frames
ret, frame1 = cap.read()
ret, frame2 = cap.read()

while cap.isOpened():
    # Compute absolute difference between frames
    diff = cv2.absdiff(frame1, frame2)

    # Convert to grayscale
    gray = cv2.cvtColor(diff, cv2.COLOR_BGR2GRAY)
```

```
# Apply Gaussian blur to reduce noise
gray = cv2.GaussianBlur(gray, (5, 5), 0)

# Apply threshold to detect motion
_, thresh = cv2.threshold(gray, 25, 255, cv2.THRESH_BINARY)

# Find contours (regions of motion)
contours, _ = cv2.findContours(thresh, cv2.RETR_EXTERNAL, cv2.CHAIN_APPROX_SIMPLE)

# Draw bounding boxes around moving objects
for contour in contours:
    if cv2.contourArea(contour) < 500: # filter small noise
        continue
    (x, y, w, h) = cv2.boundingRect(contour)
    cv2.rectangle(frame1, (x, y), (x+w, y+h), (0, 255, 0), 2)

# Resize and display
display = cv2.resize(frame1, (800, 600))
cv2.imshow("Motion Segmentation", display)

# Write the frame to output video
out.write(frame1)

# Update frames
frame1 = frame2
```



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```
# Draw bounding boxes around moving objects
for contour in contours:
    if cv2.contourArea(contour) < 500: # filter small noise
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    cv2.rectangle(frame1, (x, y), (x+w, y+h), (0, 255, 0), 2)

# Resize and display
display = cv2.resize(frame1, (800, 600))
cv2.imshow("Motion Segmentation", display)

# Write the frame to output video
out.write(frame1)

# Update frames
frame1 = frame2
ret, frame2 = cap.read()
if not ret or cv2.waitKey(30) & 0xFF == ord('q'):
    break

# Release resources
cap.release()
out.release()
cv2.destroyAllWindows()
```

Output:



Conclusion/Inferences:

By combining frame differencing, preprocessing, thresholding, and contour detection, moving objects in video streams can be reliably detected and segmented. This method provides an



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efficient and effective framework for analysing motion in videos for various real-world applications.